

National University of Computer and Emerging Sciences, Lahore Campus



Course: Digital Signal Processing
Program: BS(EE)
Duration: 3 hours
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Section: EE
Exam: Final

Course Code: EE302
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Total Marks: 100
Weight:
Page(s): 2

Student : Name: [REDACTED]

Roll No [REDACTED]

Instruction/Notes:

- In case of more than one attempts to a question, only one will be evaluated, whichever is found first.
- Answer all the questions on the answer sheet
- Calculators are allowed

Question # 1

CLO-07 [5+5+5+3=18 Marks]

Consider the filter

$$y(n] = 0.9y[n - 1] + bx[n]$$

- Determine b such that the frequency response of the filter satisfies the condition: $|H(0)| = 1$
Hint: Find $H(\omega)$ where $H(\omega) = \frac{Y(\omega)}{X(\omega)}$ (Use Fourier transform properties)
- Draw pole zero plot of the system function $H(z)$
- Determine the frequency at which $|H(\omega)| = \frac{1}{\sqrt{2}}$
- Would you classify this filter as low pass, high pass, or bandpass? Is this filter stable? Justify your answer.

$$X(\omega) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$$

Question # 2

CLO-06 [10+10=20 Marks]

- a) Find all possible $x[n]$ associated with the Z-transform:

$$X(z) = \frac{5z^{-1}}{(3 - z^{-1})(1 - 2z^{-1})}$$

- b) Find $X(z)$ and the region of convergence of each of the following sequence:

i. $x[n] = 2^n u[-n]$

ii. $x[n] = u[n + 10] - u[n + 5]$

$$\frac{1}{1-A}$$

$$\frac{1}{1-a^2}$$

$$a^n \quad |z| > a$$

Question # 3

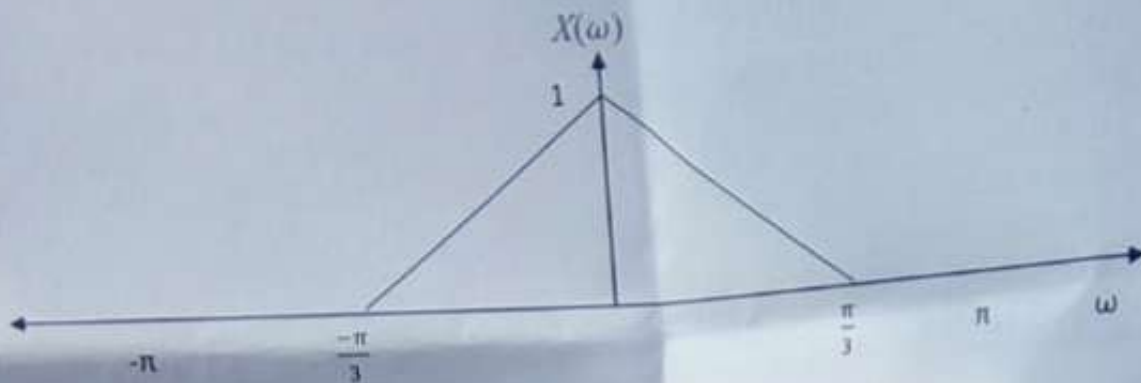
CLO-03 [10+10=20 Marks]

- a) A sequence $x[n]$ has zero phase DTFT $X(\omega)$ as sketched in figure below.



$$e^{j\omega n}$$

Formula =



Sketch the DTFT of the sequences:

- $x(n)e^{-j\pi n/3}$
- $x(n) \cos(\frac{\pi}{6}n)$

b) Find the Discrete time Fourier transform (DTFT) of the two sided sequence:

$$x(n) = \left(\frac{1}{4}\right)^{|n|}$$

Question # 4

CLO-05 [10+25=35 Marks]

a) Consider the sequence $x(n) = \delta(n) + 2\delta(n-2) + \delta(n-3)$

i. Find the 4-point DFT of $x(n)$

ii. If $y(n)$ is the 4-point circular convolution of $x(n)$ with itself, find $y(n)$ and the 4-point DFT

$$Y_4(k)$$

$$y(n) = (5, 4, 5, 2)$$

$$Y_4(k) = [16, -2j, 4, 2j]$$

$$X(k) = [3, -1+j, 2, -1-j]$$

$$X(k) = \sum_{n=0}^{N-1} x(n) \omega_N^{kn} = \sum_{n=0}^3 x(n) e^{j\frac{2\pi kn}{4}}$$

b) Consider the 4-point DFT of the signal $x(n)$ ($x(n)$ is of length 4 as well) as follows.

$$X(k) = \sum_{n=0}^3 x(n) e^{-j\frac{2\pi kn}{4}}, k = 0, \dots, 3 \quad (1)$$

$$X(k) = [1-j, 1+j, 1+j, -5+j]$$

i. What is the total complexity of the computation of $X(k)$ using equation (1)

ii. Now consider the radix-2 decimation in frequency Fast Fourier transform (FFT) implementation of $X(k)$. If $x(n) = [1, 0, -2, 3]$, draw the corresponding data flow diagram of 4-point FFT showing each stage and calculate $X(k)$ using the FFT diagram.

iii) What is the total complexity of the radix-2 decimation in time FFT algorithm? Compare it with your answer in part (a)

Question # 5

CLO-04 [7 Marks]

A sequence $x(n)$ corresponds to samples of a bandlimited signal using a sampling frequency $F_s = 10\text{KHz}$. However, the sequence should have been sampled using a sampling frequency $F_s = 12\text{KHz}$. Design a system (values of L and M) for digitally changing the sampling rate.

$$\frac{L}{M} = \frac{6}{5}$$